

Possibility and Potenzial Intervention Targets of Saffron Extract in the Treatment of Atopic Dermatitis: A Review

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ABSTRACT

Atopic dermatitis (AD) is a chronic, recurrent inflammatory skin disorder characterized by dry skin, eczema-like lesions, and severe itching. The multifaceted etiology of AD, which is not yet fully understood, includes genetic predispositions, immune dysfunctions (such as an impaired skin barrier and abnormal immune regulation), imbalances in the skin microbiota, and environmental factors, among others. In the field of AD treatment, the combination of traditional Chinese medicine and modern medicine is becoming an emerging trend. Given the potential side effects and reduced efficacy of conventional therapeutic drugs, Chinese herbal medicines offer patients new treatment options because of their unique efficacy and low toxicity. Some saffron extracts derived from saffron and gardenia, such as crocin, crocetin, and safranal, have shown promising potential in the treatment of AD. These natural ingredients not only possess anti-inflammatory and immunomodulatory properties similar to those of traditional Chinese medicines but also demonstrate excellent effects in promoting the repair of damaged skin barriers. Therefore, this article reviews the therapeutic potential of saffron extract in the treatment of AD, with a special focus on its mechanisms and potential interventions, while emphasizing the importance of herbal medicines as alternatives to traditional treatments, providing AD patients with safer and more effective treatment options.

Introduction

Atopic dermatitis (AD), a prevalent inflammatory and pruritic dermatologic condition, significantly impairs patients' quality of life [1]. Pathophysiologically, AD is characterized by disruptions in the local cutaneous microenvironment, manifesting as edema in the epidermal intercellular spaces and infiltration of inflammatory cells into the dermis [2,3]. The lifetime prevalence of AD is reported to be as high as 20% [4], and it frequently coexists with other allergic conditions such as asthma

and allergic rhinitis, leading to its recognition as a systemic disorder [5].

Traditional Chinese medicine (TCM) is well known for its potent anti-inflammatory and antiallergic properties and has been used to manage various dermatologic conditions, including AD. Evidence from eight high-quality randomized placebo-controlled trials suggests that Chinese herbal medicines can improve the size

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and severity of skin lesions and enhance sleep quality in AD patients [6]. The compound TCM dermatitis ointment (CTCMDO), which contains a mixture of six herbs—*Coptis chinensis* Franch. (family: Ranunculaceae), *Phellodendron chinense* Shneid. (family: Rutaceae), *Angelica sinensis* (Oliv.) Diels (family: Apiaceae), *Rehmannia glutinosa* Libosch. (family: Scrophulariaceae), *Curcuma longa* L. (family: Zingiberaceae), and sesame oil extracted from *Sesamum indicum* L. (family: Pedaliaceae)—has demonstrated efficacy in diminishing inflammatory responses, modulating itch-related biomolecules, and alleviating pathological changes in AD model mice [7]. *Indigo naturalis*, another TCM entity commonly used to treat autoimmune and inflammatory dermatoses such as psoriasis and allergic contact dermatitis, contains indirubin, an active component known to modulate immune responses. Indirubin reduces serum IgE levels and cytokine production and regulates Th1- and Th2-mediated immune responses [8]. Furthermore, experimental studies suggest that *Indigo Pulverata Levis* extract can inhibit the expression of inflammatory cytokines and may be a promising candidate for AD treatment [9]. These examples highlight the potential of TCM in treating skin diseases.

Saffron (botanical name: *Crocus sativus* L., family: Iridaceae) is a medicinal plant that originated in the Middle East and was later introduced to Mediterranean regions. It is widely cultivated in Iran, India, Greece, Spain, and Italy, where its cultivation has long-standing traditions and yields high-quality products [10–12]. The key bioactive components of saffron include crocin, crocein, safranal, and other medicinally significant compounds [13]. Crocin (a glycoside derivative of crocetin) compounds contribute to the color intensity of saffron, whereas safranal (a monoterpene aldehyde) provides its typical flavor [14].

Gardenia (botanical name: *Gardenia jasminoides* Ellis., family: Rubiaceae) is a dried and ripe fruit native to Asia, including China, India, Japan, and Thailand, and later spread to Africa and Oceania [15,16]. The primary bioactive ingredients of gardenia include genipin, gardenoside, crocin, iridoids, and other compounds [17,18]. Among these, crocin has attracted increasing attention from researchers. Compared with similar compounds in saffron, crocin extracted from gardenia and its derivatives has lower toxicity, reduced risk of allergic reactions, and greater environmental value [19,20].

Saffron extract, produced from a combination of saffron and gardenia—widely used in TCM and as food [21]—contains a diverse range of chemical constituents, including anthocyanins, flavonoids, and carotenoids including crocin, crocetin, and safranal [22]. These compounds exhibit various pharmacological activities, including anti-inflammatory [23], antioxidant [24], antibacterial [25], and immunomodulatory [26] effects. Furthermore, prior research has highlighted the therapeutic benefits of these substances across multiple physiological systems, including the nervous [27], endocrine [28], reproductive [29], cardiovascular [30], and digestive [31] systems.

This study aims to elucidate the comprehensive impact of saffron extract on the development and progression of AD, summarize its potential therapeutic targets, and provide novel insights into the understanding and treatment of this condition.

Literature Search Strategy

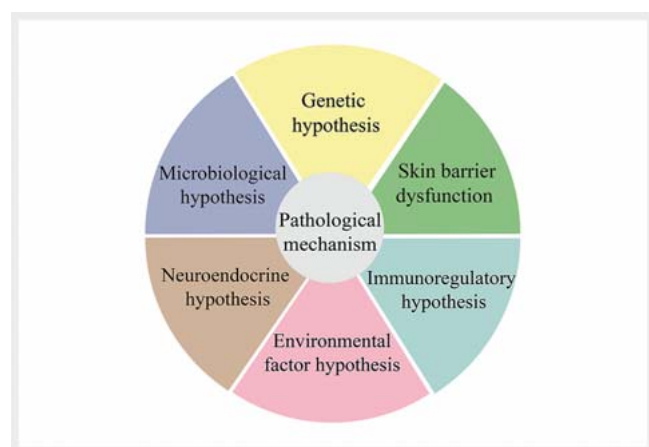
The purpose of this review is to explore the potential of saffron extract in the treatment of AD, with a focus on its mechanisms of action and safety profiles. This review also aims to evaluate the potential applications of saffron extract, particularly crocin and crocetin, in managing AD, with the goal of providing safer and more effective treatment options for AD patients. Relevant information from the past 10 years was gathered through online search engines and scientific databases such as Google Scholar, PubMed, Web of Science, ScienceDirect, Scopus, and Wiley Online Library. The main topics covered in the search included phytochemistry, atopic dermatitis, regulatory mechanisms, immunomodulation, and safety. The following search terms were used: saffron, *Crocus sativus* L., gardenia, *Gardenia jasminoides* Ellis., saffron extract, crocin, crocetin, safranal, atopic dermatitis, clinical trials, safety, and the Boolean operators “AND” and “PLUS”.

Pathological Mechanism of Atopic Dermatitis

The pathological mechanism of AD is understood to involve a complex, sequential process. Several hypotheses have been proposed to explain the underlying pathophysiology of AD, including the genetic hypothesis, skin barrier dysfunction hypothesis, immunoregulatory theory, environmental factor hypothesis, neuroendocrine hypothesis, and microbial hypothesis (► Fig. 1).

The strong familial prevalence of AD supports the genetic hypothesis, highlighting the critical role of genetic factors as significant risk determinants for the disease [32]. Recent studies have revealed a robust correlation between the onset of AD and genetic variants categorized into three primary domains. First, genes related to skin barrier function, such as *FLG*, *LOR*, *IVL*, *SPINK5*, and *TMEM79*, are essential for keratinocyte differentiation and maintaining skin barrier integrity [33–37]. Second, immune-related genes involved in both innate and adaptive immunity contribute to disease development. These include genes encoding Th2 cytokines and various chemokines, such as TSLP and OX40L [38–40], as well as non-Th1 and non-Th2 cytokines [41,42] and genes encoding immunomodulatory cytokines such as *NOD1*, *NOD2*, *CCL3*, *CXCL9*, *CXCL10*, *CXCL11*, and *IL5RA* [43–46]. Third, epigenetic factors, including DNA methylation, histone modifications, and microRNAs, have been implicated in AD susceptibility [47–54].

The skin barrier dysfunction hypothesis posits that the skin, as the body's primary defense against external threats, plays a crucial role in the development of AD. Disruptions in the skin barrier facilitate the entry of allergens, irritants, and pathogens, triggering both local and systemic immune reactions [55]. Specifically, skin barrier dysfunction and overactive type II immune responses drive the clinical manifestations and pathology of AD. This abnormal barrier is influenced by a combination of genetic, immune, and environmental factors that affect gene expression, structural proteins, and lipid profiles [56]. Loss-of-function mutations in the filaggrin (*FLG*) gene are the strongest known genetic risk factor for AD [57]. *FLG*, encoded by the *FLG* gene, is vital for maintaining skin barrier function and moisture levels. Deficiency of *FLG* results in a compromised barrier, increased water loss, and dryness,



► **Fig. 1** Six major hypotheses on the pathogenesis of atopic dermatitis.

which can disrupt the skin microbiome and exacerbate pruritus [58].

The immunoregulatory hypothesis emphasizes the central role of immune system dysregulation in AD progression, particularly the imbalance between Th2 and Th1 responses and aberrant expression of related cytokines and immunomodulatory molecules. In the early stages of AD, Th2 cells overproduce cytokines, leading to excessive IgE production and eosinophil activation—hallmarks of allergic responses. As the disease progresses, Th1 immune responses may become involved, particularly in the chronic phase [59]. This immune imbalance not only triggers skin inflammation but also exacerbates skin barrier dysfunction, creating a vicious cycle. During this process, IFN- γ and IL-4 regulate the differentiation of Th0 cells into Th1 and Th2 cells, respectively. IRF-1 and IRF-2, which are induced by IFN- γ , bind to three different IL-4 promoter sites, where they act as transcriptional repressors and inhibit IL-4 gene transcription [60]. Conversely, IL-4 produced by Th2 cells activates GATA-3, promoting IL-4 gene transcription and suppressing IFN- γ gene transcription, thus establishing mutual regulation between Th1 and Th2 [61]. In AD, Th2-mediated immune responses generally dominate Th1 responses, leading to an imbalance between the two [62]. Th2 activation stimulates B cells and T cells to produce large amounts of IgE, further sensitizing the immune system and inducing the release of inflammatory mediators such as leukotrienes and histamine by mast cells, thereby exacerbating AD [63].

The environmental factor hypothesis highlights the significant role of external factors in the progression of AD. Studies have shown that airborne allergens (e.g., dust mites [64], pollen [65]), environmental pollutants [66], climate changes (e.g., dry or extreme temperatures) [67], chemical and skin irritants, and microbial exposures can trigger or worsen AD symptoms. These factors may affect skin barrier function, alter immune responses, or influence other biochemical pathways, collectively contributing to the development and exacerbation of AD.

The neuroendocrine hypothesis suggests that changes in the neuroendocrine system can directly or indirectly affect skin func-

tion and immune responses. Emotional stress and psychological factors can influence skin inflammation and pruritus by modulating neurotransmitters (e.g., neuropeptides, substance P) and hormones (e.g., cortisol) [68]. These neuroendocrine changes may also lead to immune system dysregulation, disrupting the Th1/Th2 balance and worsening AD symptoms [69]. This hypothesis underscores the importance of the psychoneuroendocrine-skin axis in AD, illustrating the interplay between psychosocial factors and neuroendocrine mechanisms in the onset and progression of the disease [70, 71].

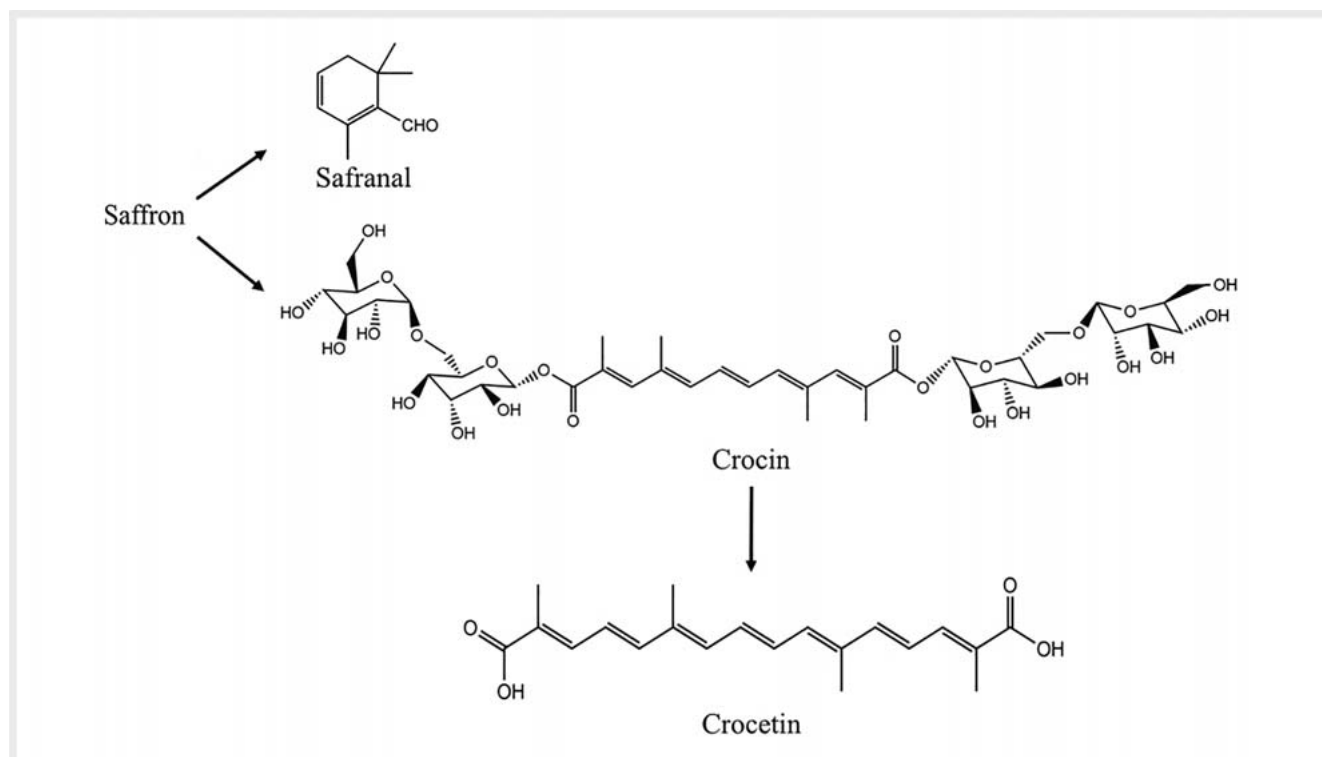
The microbial hypothesis of AD primarily focuses on the role of the skin microbiome in disease development. Bacteria posits that bacteria such as *Staphylococcus aureus* (*S. aureus*) can invade the compromised skin barrier, release toxins, and induce inflammatory responses, exacerbating skin inflammation and pruritus [72, 73]. Furthermore, microbial dysbiosis may affect skin immunity, leading to overreactions or inappropriate immune responses to other pathogens. This hypothesis highlights the importance of maintaining a balanced skin microbiome for preventing and treating AD, suggesting that microbial interventions could be effective strategies for managing this disease [74].

The skin is home to a highly diverse symbiotic community of more than 1000 bacterial species from 19 different phyla [75]. According to the microbial hypothesis, disturbances in the skin and gut microbiomes are linked to AD development [76]. For example, *S. aureus* can invade damaged skin barriers and secrete various pathogenic substances that induce inflammatory responses [77]. Additionally, the gut microbiota can influence skin immunity through intestinal immune mechanisms [78]. Studies have shown that the intestinal flora of AD patients differs significantly from that of healthy individuals, characterized by reduced microbial diversity and increased abundances of *Escherichia coli* (*E. coli*) and *S. aureus*, whereas beneficial microorganisms such as *Lactobacillus* and *Bifidobacterium* are diminished [79–81]. Factors such as breastfeeding methods, environmental conditions, lifestyle habits, and probiotic use can all impact the expression and composition of the intestinal microbiota [82, 83]. Moreover, the intestinal microbiota influences the proliferation of various immune cells, including innate lymphoid cells, Th17 cells, Th1 cells, Th2 cells, and regulatory T cells [84–86]. Disruptions in the gut flora can impair oral tolerance and reduce intestinal epithelial integrity, leading to increased allergen exposure and heightened allergic susceptibility. This involves the activation of gut-associated lymphoid tissue, cytokine release, and increased intestinal permeability [87]. Thus, preserving the balance of the gut microbiome may be a promising strategy for preventing and treating AD.

The complex nature of AD indicates that it is influenced by a wide array of factors rather than a single cause. This multifactorial aspect necessitates a comprehensive approach to treatment, addressing the various contributing elements.

Properties of the Saffron Extract

Crocin, crocetin, and safranal are significant pharmacological compounds, some of which can be extracted from saffron and gardenia (► **Table 1**). Additionally, saffron extract contains a variety of other beneficial components, including vitamins, minerals,



► **Fig. 2** The chemical structure of saffron and its key bioactive constituents: safranal, crocin, and crocetin.

monoterpenes, anthocyanins, carotenoids, and flavonoid compounds [102].

Crocine, a water-soluble carotenoid and the primary constituent of saffron, exists mainly in all-trans and 1,3-cis configurations. It is converted into crocetin upon deglycosylation, a process that allows rapid absorption in the intestine and ensures its stability in the body. Safranal, another key component of saffron, is not directly derived from crocine but is produced from picrocrocine during the drying and degradation processes. Known for its distinct aroma, safranal is one of the main volatile compounds of saffron. These components, especially crocine and safranal, are recognized for their antioxidant and anti-inflammatory properties [103] (► **Fig. 2**).

Furthermore, crocine and crocetin have shown potential in the treatment of various human diseases, including diabetes, depressive disorder, and coronary artery disease, owing to their potent antioxidant and anti-inflammatory effects (► **Table 2**). However, while these compounds have demonstrated promising therapeutic benefits in multiple conditions, there is currently a lack of robust clinical evidence supporting their effectiveness specifically in the treatment of AD. Further research is needed to explore the potential of saffron extracts in this context.

Regulation of Th2 Cytokines and the JAK Pathway

Cytokines significantly contribute to the pathogenesis of AD, indicating that they are primary targets for therapeutic intervention. Specifically, IL-4 and IL-13 bind to the type II receptor complex, which consists of IL-4R α and IL-13R α 1. This interaction activates the protein kinases JAK1, JAK2, JAK3, and Tyk2, leading to the phosphorylation of STAT6/STAT3 [113]. This pathway is pivotal in AD as it mediates a range of biological responses, including cell proliferation, differentiation, migration, apoptosis, and immune regulation.

Research indicates that JAK inhibitors can modulate several signal transduction pathways related to inflammatory and immunological responses, highlighting their effectiveness in treating AD through topical or oral administration [114]. The 2020 edition of the Chinese recommendations for the diagnosis and management of AD acknowledges the effectiveness of the biological JAK inhibitor baricitinib in treating moderate to severe AD in adults. It also recognizes oral upadacitinib and tofacitinib as potential systemic treatments [1].

JAK inhibitors suppress key cytokines involved in AD pathogenesis by inhibiting the JAK/STAT signaling pathway. During a 16-week treatment period with baricitinib, most patients reached the primary endpoint, scoring 0 (clear) or 1 (almost clear) on the validated Investigator Global Assessment (iGA). Inflammatory signs of AD significantly improved or completely resolved. Addi-

► **Table 1** Methods for extracting crocin, crocetin, and safranal from saffron and gardenia.

From	Ingredients	Extraction method	Results	References
Saffron	Crocine	Ultrasonic-assisted extraction	The optimal methanol concentration, sonication time, and duty cycles for achieving the highest concentration of crocin were 50%, 30 min, and 0.2 s on/0.8 s off, respectively.	[88]
	Crocine, Crocetin	Aqueous two-phase system (ATPS)	The optimal conditions for ATPS were a volume ratio of 5:3.2, 19.8% (w/w) ethanol, 16.5% (w/w) potassium phosphate, a tie-line length (TLL) of 25% (w/w), 0.1 M NaCl, and a sample load of 2% (w/w).	[89]
	Crocine, Safranal	Solid-phase extraction	The best solvent for binding was 2% (v/v) acetic acid, and the best washing solvent was acetonitrile (ACN).	[90]
	Crocetin	Subcritical water extraction (SWE)	Response surface methodology was used to optimize the subcritical water extraction of saffron compounds, and the compounds were characterized by GC-MS and HPLC. The optimal conditions for the process were found to be an extraction temperature of 105 °C and an extraction time of 7.32 min.	[91]
	Safranal	Supercritical CO ₂ extraction	The optimal conditions for crocin extraction were 44 °C, 19.3 MPa, a flow rate of 1.0 cm ³ /min, and an extraction time of 110.0 min that resulted in a 33% (w/w) recovery; For safranal extraction were 92 °C, 21.3 MPa, 0.9 cm ³ /min and 122 min that resulted in a 90% (w/w) recovery.	[92]
	Safranal, Crocetin	Microwave-assisted extraction (MAE)	MAE was on the whole more successful than SWE in extracting the crocetin esters. The optimal extraction temperature was 95.91 °C, the MAE time was 30 min, and the ethanol concentration was 59.59% (v/v).	[93]
	Saffron apocarotenoids	Ultrasound-assisted extraction	The optimal solvent, extraction time, and saffron concentration for achieving the highest concentration of safranal were chloroform, 15 min, and 20 g/L of saffron, respectively.	[94]
	Crocine	Supercritical fluid extraction (SFE)	A mixture of CO ₂ and water or methanol was used as a modifier to extract crocetin from safranal. The optimal extraction conditions for crocetin were 30 MPa and 80 °C, while the optimal extraction conditions for crocin aldehyde were 40 MPa and 80 °C.	[95]
	Crocine, Crocetin	Pulsed electric field (PEF)	Field strength, pulse number, and pulse width all improved the recovery rate of saffron carotenoids. The best combination of these conditions was 5 kV, 35 µs and 100 µs, respectively.	[96]
Gardenia	Crocine	Macroporous resins	The resins with the highest adsorptive capacity and selectivity for crocin were XAD1180, HP20, HPD-100A, and AB8 (with AB8 being the best); based on static absorption/desorption experiments, the best resin was LX60.	[97,98]
	Crocine, Crocetin	Pulse-ultrasonication-assisted extraction (P-UAE)	The optimized P-UAE conditions (extraction time 120 s, ultrasonic power 400 W, solvent-to-solid ratio 40 mL/g, extraction temperature 35 °C) led to the maximum yield of crocins (36.97 mg/g DW).	[99]
	Crocine, Safranal	Foam-mat drying	The optimal extraction parameters were extraction temperature of 55 °C, time of 57 min, 24% fruit in solvent, 56% ethanol concentration. The crocin content of gardenia powder remained good (6.64 mg/g) and low water activity, with moisture levels of 0.33% and 5.72%, suitable for storage.	[100]
	Crocetin	Organic solvent extraction	Methanol and deionized water were appropriate solvents for pigment recovery with maximum yields of at least 17% from the floral tissue.	[101]

tionally, patients experienced improvement in itching symptoms within 1 to 2 weeks, and other aspects, such as skin pain, nocturnal awakenings, and quality of life scores, improved within a week [115,116]. Mometinib, a novel JAK1/JAK2 inhibitor, suppresses the phosphorylation of STAT1, STAT3, and STAT5 in lesion areas, consequently reducing the expression of inflammatory factors such as IL-4, IL-5, IFN-γ, and the total serum IgE level in DNCB-in-

duced AD mice. It also leads to reductions in histological measures like epidermal thickness and mast cell count in the lesion area [117].

Multiple studies have confirmed the ability of saffron extract to interfere with the JAK/STAT signaling pathway. It can inhibit IL-6-induced STAT3 activity and gene expression in hepatocellular carcinoma cells and block the activity of nonreceptor tyrosine ki-

► **Table 2** Therapeutic effects and side effects of crocin and crocetin.

Ingredi- ents	Main objec- tives	Method	Therapeutic effect	Side effect	Refer- ences
Crocine	Diabetes	30 mg/day crocin for 3 months	Controlling the level of FBS and HbA1c in patients.	No severe and specific side effects	[104]
	Metabolic syndrome	30 mg/day crocin for 8 weeks	Increased the serum cholesteryl ester transfer protein in patients (not significant).	None mentioned	[105]
		100 mg/day crocin tablets for 6 weeks	Significant reductions from baseline measurements in the levels of total cholesterol ($P < 0.001$) and triglyceride ($P = 0.003$).	No complication and well tolerated	[106]
	Diabetic macular edema (DME)	5 mg/day or 15 mg/day crocin tablets for 3 months	5 mg/day crocin tablet could clinically improve HbA1c, FBS, CMT, and BCVA (not significant).	No severe and specific side effects	[107]
	Depressive disorder	SSRI drug plus 30 mg/day crocin tablets for 4 weeks	Improved scores on BDI, BAI, and GHQ compared to placebo group (P value < 0.0001).	No severe and specific side effects	[108]
	Osteoarthritis (OA)	Krocina tablets (contain 15 mg of crocin with a purity of $> 90\%$) for 4 months	Gene expression levels of microRNA-21 and microRNA-155 were significantly decreased and increased, respectively.	No side effects	[109]
Crocetin	Coronary artery disease	10 mg/day crocetin for 2 months	Decreased levels of Hcy, h-FABP, intercellular adhesion molecule 1, vascular cell adhesion molecule 1, monocyte chemoattractant protein 1, systolic and diastolic blood pressures; significantly improved levels of HDL.	No serious side effects	[110]
	Sleep	Capsule contained 7.5 mg of crocetin for two intervention periods of 2 weeks each	Significantly reduced the number of waking episodes; improved the quality of sleep.	No side effects	[111]
	Myopia control in children	Capsule contained 7.5 mg of crocetin for 24 weeks	Against myopia progression and axial elongation in children.	No adverse effects	[112]

nases, including JAK1, JAK2, TYK2, and c-Src kinases, in Hep3B cells. Consequently, this inhibits STAT3 phosphorylation and disrupts the JAK/STAT signaling pathway [118]. Additionally, crocin can hinder PDGF-BB-induced VSMC proliferation and phenotypic transition, thus mitigating atherosclerotic plaque formation. PDGF-BB significantly elevates the phosphorylation levels of JAK1, JAK2, and STAT3 in VSMCs, but crocin can inhibit this effect in a concentration-dependent manner [119]. Research also indicates that safranal can protect against intestinal injury induced by JAK/STAT signaling in *Drosophila* by inhibiting the Ecc15 signaling pathway [120] (► **Fig. 3**).

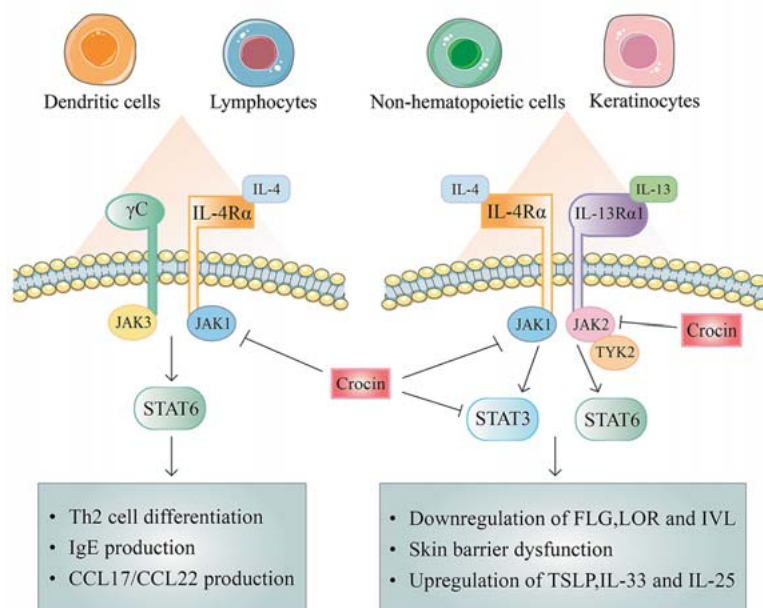
Like crocetin, baicalin is a natural flavonoid that functions as a JAK inhibitor, blocking the JAK/STAT signaling pathway and disrupting cytokine secretion and function. This makes it suitable for treating AD-related skin diseases [121], suggesting that crocetin may also serve as a JAK/STAT inhibitor in AD treatment.

Doxorubicin injection acts as an IL-4R antagonist that specifically targets type II inflammatory diseases. It is distinguished as the first and only targeted biological agent approved for treating moderate and severe AD in adults [122]. Saffron extract may have

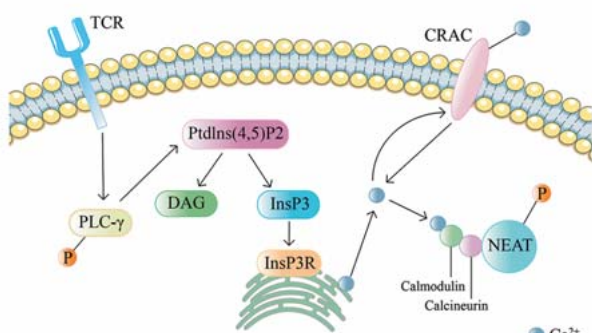
a similar regulatory effect and may interfere with AD through this mechanism. Studies have demonstrated that crocin can significantly reduce IL-4 and IL-13 levels in the lungs of mice, improve oxidative stress, alleviate angioedema, and notably reduce the number of inflammatory cells around blood vessels and bronchi. It also markedly decreases the number of macrophages and inflammatory cells in the alveolar cavity by less than 25%. Furthermore, saffron glycoside can modify immune responses by lowering the ratios of T-bet to GATA-3 and INF- γ to IL-4, reducing inflammation and increasing the Th1/Th2 ratio [123, 124].

Regulation of Other Cytokines

Th17 cells and Th22 cells, along with their associated cytokines such as IL-23, IL-17, IL-22, IL-26, and IL-31, are also crucial in the development and progression of AD. These immune system components have been identified as significant contributors to the underlying mechanisms driving the disease [125, 126] (► **Table 3**).



► **Fig. 3** Crocin regulates Th2 cytokines and the JAK pathway. Hematopoietic cells that express the IL-4Rα/γC complex include dendritic cells and lymphocytes. Binding to IL-4Rα/γC and IL-4 but not IL-13 activates the downstream signaling pathways JAK1/JAK3 and subsequently STAT6. Th2-mediated T-cell differentiation, IgE synthesis in B cells, and Th2 chemokine production from dendritic cells, including CCL17 and CCL22, are all induced by activation of the IL-4–IL-4Rα/γC–JAK1/JAK3–STAT6 axis. On the other hand, keratinocytes and other non-hematopoietic cells express the IL-4Rα/IL-13Rα1 complex. IL-13 and IL-4 bind to IL-4Rα/IL-13Rα1 and activate JAK1/TYK2/JAK2 downstream, followed by STAT6/STAT3. The IL-13/IL-4–IL-4Rα/IL-13Rα1–JAK1/TYK2/JAK2–STAT6/STAT3 axis is activated, which results in the downregulation of FLG, LOR, and IVL expression. Additionally, it inhibits the nuclear translocation of OVO-like 1 (OVOL1), compromises the integrity of the skin barrier, and increases keratinocyte thymic stromal lymphopoietin (TSLP), IL-25, and IL-33 production. Crocin can inhibit STAT3, JAK1, and JAK2 to intervene in the progression of this pathway.



► **Fig. 4** Figure of calcium signaling pathway. Relevant receptors in the calcium signaling pathway induce NFAT activation. The T cell receptor (TCR) is linked to ligand recognition and activates receptor-associated tyrosine kinases, subsequently initiating the calcium-calmodulin-calcineurin-NFAT pathway. (CRAC, Ca²⁺-Release Activated Ca²⁺; DAG, Diacylglycerol; InsP₃, Inositol triphosphate; InsP₃R, Inositol triphosphate receptor; P, Phosphate groups; PLC-γ, Phospholipase C-γ; PtdIns(4,5)P₂, Phosphatidylinositol 4,5-bisphosphate; TCR, T cell receptor; NFAT, Nuclear Factors of activated T cells.)

Regulation of the ERK Pathway

The ERK pathway is one of the MAPK signaling pathways, and its activation promotes the expression of matrix metalloproteinases (MMPs). These proteolytic enzymes are secreted by proinflammatory cells and uterine placental cells. The concentration of MMP-2 in serum can serve as a biological marker for AD, which often disrupts the skin barrier [133]. Saffron extract has been shown to increase the levels of type I and type III collagen while downregulating the expression of ERK1/2 and MMP-2 [134]. In a rat model of brain ischemia, crocin reduced the activity of NADPH oxidase and MMP-2 in brain tissues, acting as an antioxidant to alleviate stroke. Studies have confirmed that crocin modulates MMP-2 by regulating ERK1/2 and mitigating oxidative stress [135].

Inhibition of Calcineurin and NFAT Expression

Calcineurin (CaN) is a critical enzyme in human immunoregulation that primarily targets the NFAT family of proteins, which includes NFAT1~5. NFAT1 is associated with promoting Th2 cell responses and increasing IL-4 expression while inhibiting IFN-γ production. Additionally, the cooperative action of NFAT1 and NFAT4 can increase TCR reactivity, potentially leading to lymphoproliferative diseases and intensifying Th2 cell responses. In contrast,

► **Table 3** Functions and roles of cytokines.

Cytokine	Function	Role in AD	
IL-23	Involves activating Th17, which produce active cytokines (IL-17A, IL-17F, TNF, IL-6); Involves the phosphorylation of receptor-associated JAK and specific tyrosine residues.	Increased inflammation; Increased amyloid deposition; Possible neuronal damage.	[127]
IL-17	Involves the ROR- γ T/STAT3 transcription factor activation; Activates downstream pathways such as NF- κ B, the C/EBP family, and MAPK, leading to the expression of downstream genes.	Involved in inflammation and immune regulation systems.	[128, 129]
IL-22	Targets non-hematopoietic cells (e.g., epithelial cells); Involved in keratinocyte proliferation.	Serum levels correlate with AD activity; Frequency of CD4+IL-22+ T cells decreases, while CD8+ T cells show increased IL-22 levels.	[130]
IL-26	Bridges Th17 and Th2 cell responses; Enhances secretion of TSLP, CXCL1, and CCL20 via JAK/STAT signaling pathway.	Exacerbates AD-like skin inflammation; Increases expression of Th2-type cytokines such as IL-13.	[131]
IL-31	Upregulates Nppb gene expression, inducing brain natriuretic peptide (BNP) release.	Promoting inflammation; Exacerbating itching.	[132]

NFAT2 inhibits Th2 cell responses and IL-4 expression [136] (► **Fig. 4**).

Patients with AD may find relief through the use of topical calcineurin inhibitors (TCIs) like tacrolimus, pimecrolimus, and glucocorticoids for localized anti-inflammatory treatment of skin lesions. However, the local use of TCIs and glucocorticoids may result in adverse reactions, such as skin burning [137], itching [138], atrophy [139], and Cushing's syndrome [140]. In contrast, saffron extract has minimal negative effects on the human body and offers several advantages as a natural remedy. Studies have shown that saffron extract can increase the expression of calcineurin and NFAT4 (NFATc3) induced by oleic acid [141]. It also elevates the IFN- γ /IL-4 ratio, regulating the differentiation of Th1/Th2 cells and the production of antibodies by B cells [142]. These findings suggest its potential to counteract Th2 inflammation by modulating these processes.

Inhibition of Transglutaminase Expression

Transglutaminase 2 (TG2) plays a crucial role in the formation of the skin barrier by regulating the opening of endoplasmic reticulum calcium channels, which is directly linked to the pathophysiology of AD [143, 144]. Studies involving TG2-gene-deleted mice have shown significant reductions in inflammatory responses such as erythema, edema, and infiltration of inflammatory cells following local epidermal stimulation. These findings suggest that excessive TG2 levels may hinder the restoration of the epidermal barrier by triggering an unfavorable inflammatory response [145]. A strong positive correlation has been established between TG2 mRNA, TG2-specific IgE SCORAD scores, and peripheral blood eosinophil counts in AD patients [146]. Furthermore, the Eui Man Jeong team explored the relationship between oxidative stress and TG2 expression. They found that lower oxidative stress (1 mM H₂O₂) promoted TG2 expression in HeLa cells, while higher levels (3 mM H₂O₂) caused intracellular calcium overload and trig-

gered TG2 degradation via the calcium-mediated ubiquitin-proteasome pathway [147]. These results suggest that AD patients may experience calcium influx and elevated TG2 expression at specific oxidative stress.

Saffron extract may regulate TG2 expression through two distinct mechanisms. First, it effectively inhibits oxygen free radicals [148], demonstrating antioxidant properties that can regulate oxidative stress levels in the skin lesion microenvironment, thereby influencing TG2 expression. Second, crocin and crocetin can directly modulate calcium channel activity, including the inhibition of L-type calcium channels, leading to calcium efflux from adult rat ventricular myocytes [149, 150]. These findings suggest that saffron extract may enhance intracellular calcium regulation, down-regulate excessive TG2 expression, inhibit specific IgE production, improve the skin barrier, and, therefore, offer potential therapeutic effects in AD.

Modulating Microbiota

S. aureus, a bacterium that damages the skin barrier and secretes a cytotoxin in the form of water-soluble monomers, creates heptameric pores in the host cell membrane and can cause and aggravate dysbacteriosis of the microflora on the surface of the skin [151]. It disrupts the integrity of the epidermal barrier by directly forming pores in keratinocytes [152]. Additionally, *S. aureus* produces at least 10 proteases, some of which facilitate lysis and penetration of the stratum corneum. These proteases can also activate endogenous cell proteases such as KLK6, KLK13, and KLK14, resulting in increased serine protease activity in keratinocytes. This cascade promotes the degradation of DSG-1 and FLG, further compromising the skin barrier [153]. Various microbes, including *S. aureus*, have been detected in the skin lesions of individuals with AD. Treatment with antibiotics targeting these microbes effectively reversed the ecological imbalance and reduced inflammation in the lesions, underscoring the critical role of *S. aureus* in

driving AD pathogenesis. This highlights the microbiome–host immune axis as a promising avenue for future AD treatments [154].

Saffron extract, a natural antibiotic, shows promise in combating *S. aureus* and other microorganisms, suggesting a novel approach for antibacterial therapy. Additionally, a study revealed an enrichment of *E. coli* in the intestinal microbiota of 63 infants with AD, aged 3 weeks to 12 months [155]. Oral administration of crocin was shown to mitigate *E. coli* colonization in the intestines and alter the levels of symbiotic bacteria. Specifically, 40 mg/kg crocin increased the abundance of *Lactobacillus*, restored short-chain fatty acid (SCFA) levels in feces, and improved disrupted intestinal flora and damaged intestinal barriers in mice [156].

Discussion

Saffron extract has demonstrated significant effects in managing AD. It modulates immune responses, particularly influencing Th2 cytokines, and regulates both the skin and gut microbiota, specifically impacting the colonization of *E. coli* and *S. aureus*, which are key players in AD. These findings underscore the therapeutic potential of saffron extract as a natural treatment option for AD, offering a novel approach to managing the condition [157].

When comparing saffron extract to current AD treatments, it becomes evident that saffron could serve as a safer and natural alternative, with a lower likelihood of adverse effects. Traditional treatments, including topical corticosteroids and immunomodulators, are effective but often come with side effects like skin thinning and systemic absorption issues [158,159]. In contrast, the natural composition of saffron extract may reduce such risks. While prolonged exposure to saffron pollen may cause allergic reactions in some individuals, saffron is generally considered safe for most people [160,161]. Common side effects of saffron and its extracts include nausea, dry mouth, loss of appetite, and headaches, but there have been no reports of severe adverse reactions from saffron or crocin tablets [162–164]. Although high doses of crocin, crocetin, and safranal have been linked to embryonic malformations in animal models, and safranal exhibits higher toxicity, these compounds are generally regarded as safe at therapeutic doses [165].

Despite saffron's therapeutic potential, it remains a costly ingredient because its labor-intensive harvest from the flower's stigma results in low yields. In contrast, gardenia, which contains similar compounds such as crocin and crocetin, is more widely cultivated and easier to harvest, making it a more practical source for extraction and use in treatment [166,167]. Previous studies have shown that crocin from saffron has a bioaccessibility of 50%, lower than the 70% bioaccessibility observed in gardenia extract under gastrointestinal digestion conditions [168]. Animal studies have also indicated that crocin is rapidly converted to crocetin after oral administration, with its blood circulation is 56 to 81 times higher than the applied crocetin concentration [169]. Therefore, while crocin, crocetin, and safranal all show potential for AD treatment, crocin offers a higher utilization rate and is more suitable for practical application, followed by crocetin. Safranal, being more toxic and harder to extract, is less favored for therapeutic use.

In treating other conditions such as diabetes, osteoarthritis, and coronary artery disease, crocin and crocetin are primarily used orally with minimal side effects (► Table 2). The application of crocin in AD has been mostly topical or subcutaneous in animal models [170,171]. However, further clinical studies are needed to evaluate whether crocin can be applied topically or systemically in humans and to assess its effectiveness and safety compared with standard AD treatments. Key challenges in integrating saffron extract into clinical practice include determining the optimal dosage, understanding its pharmacological mechanisms, and ensuring consistent extract quality.

The concept of the gut–skin axis, which describes the interconnected relationship between gut and skin health, is particularly relevant in skin conditions like AD. Dysbiosis, or the alteration of the gut and skin microbiomes, is associated with immune response changes and may contribute to AD pathogenesis. Research on the role of the gut microbiota in AD is complex; studies have shown that probiotics can improve AD severity in some cases, but the results are mixed, with other studies failing to show such benefits despite changes in the gut microbial composition [172]. The relationship between the onset and severity of pre-existing AD and the gut microbiome remains a topic of ongoing debate, necessitating further investigation.

TCM has a significant effect on modulating the gut microbiota, which may have implications for the gut–skin axis and AD treatment. TCM can influence the structure, composition, function, and metabolic byproducts of the intestinal microbiota, either directly or indirectly [173]. This offers potential for integrating TCM as an adjunct therapy or personalized treatment approach for AD. Moreover, exploring the byproducts of Chinese *materia medica* and marine-derived materials in the context of TCM could broaden the range of applications for AD treatment and beyond [174,175].

Conclusion

Extracts from saffron and gardenia, including crocin, crocetin, and safranal, share comparable properties that contribute to the treatment of AD, particularly by enhancing the ability of the skin barrier to heal.

This review highlights the utility of herbal plants as alternatives to conventional treatments, which often come with undesirable side effects. Specifically, it focuses on the mechanisms through which saffron extract, especially crocin and crocetin, may intervene in AD treatment.

Future research should aim to elucidate the pathogenetic mechanisms of saffron extract in AD, exploring its molecular interactions and immune response dynamics. This could pave the way for novel, personalized treatment strategies based on these mechanisms or specific subgroups in AD therapy.

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Conflict of Interest

The authors declare that they have no conflict of interest.

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